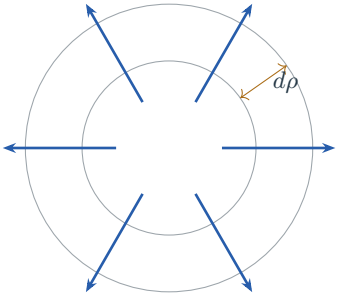
# M1-30 柱径向场要保留侧面积因子




同轴薄壳：侧面积随 $\rho$ 增大，图示 $k>0$

$$\mathbf{A} = k\rho\mathbf{e}_\rho$$

$$\nabla\cdot\mathbf{A} = \frac{1}{\rho}\frac{d(\rho A_\rho)}{d\rho} = 2k$$

$$\rho A_\rho = k\rho^2$$

外侧场更强、面积也更大，两者都要计入。